



# ClaimTrace

2026.8.19



C A T A L O G



Project Scope & Users



Targeted Market  
& Gap Analysis



Solution  
Architecture



T h e   T e a m

## The Problem

When a research team writes a paper, every claim cites a source. But nobody verifies whether the source actually says what the claim claims it says. Checking one citation means downloading a PDF, searching for the right passage, and reading it — minutes per citation, hours per paper, and rarely done before submission.

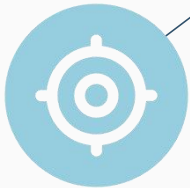
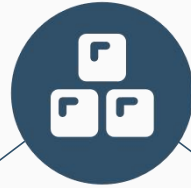




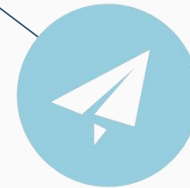
## Project Scope & Users

## Users' pain points

Verifying a citation is slow and breaks the writing flow (leave Overleaf → search → read → come back)



Co-authors' claims are hard to trust — you didn't write them, but you're responsible for them



AI tools can fabricate references that look real but don't exist

## Scope & objective

ClaimTrace is an academic citation audit tool embedded in Overleaf. Hover any `\cite{...}` and instantly see the original source passage it points to, aligned against the claim that cites it — so you catch misquoted, over-stated, or hallucinated citations while you write, not after reviewers do.



The title is centered and framed by a dark blue L-shaped line on the left and a dark blue L-shaped line on the right. Two vertical light blue bars are positioned on either side of the text.

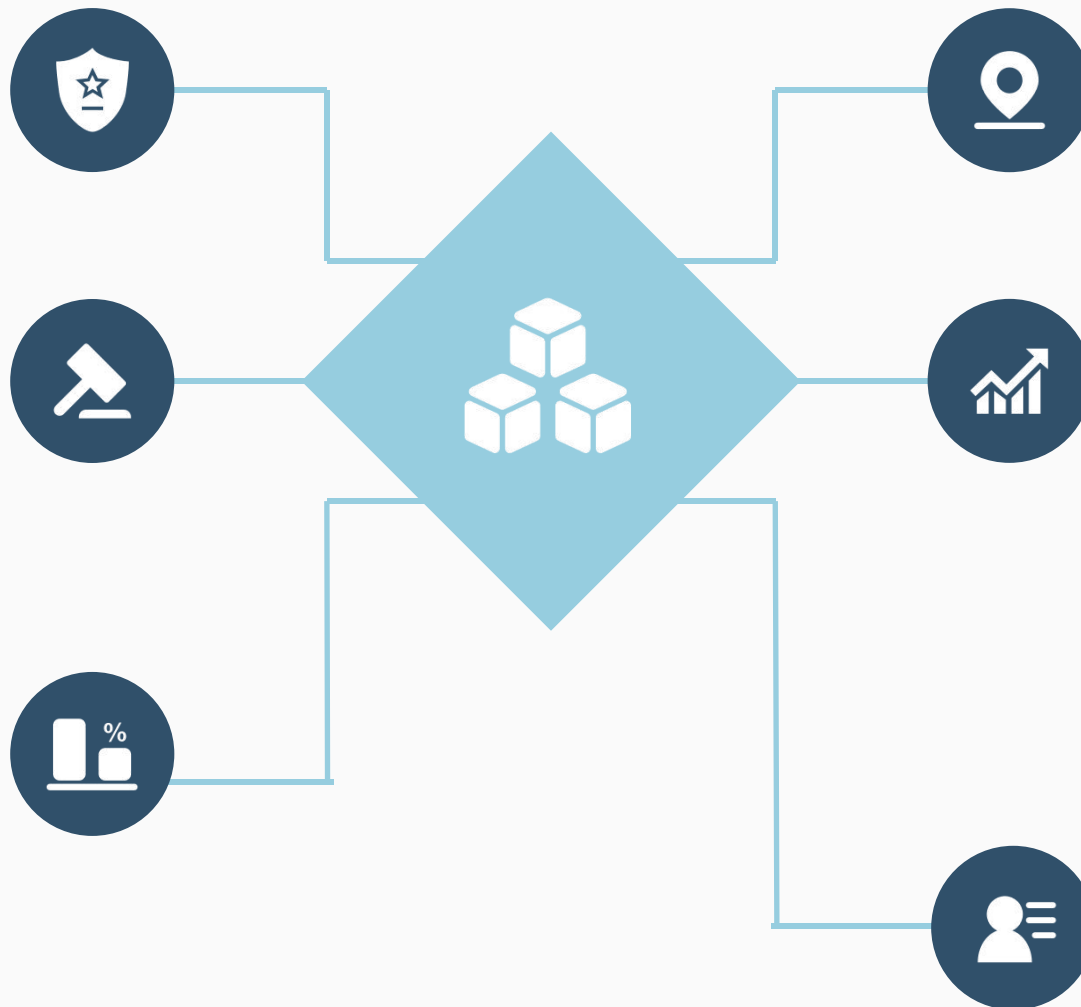
## Targeted Market & Gap Analysis

# Why this matters

An analysis covering 12,402 biomedical papers published in 2025 found that 7.1% of the papers cited at least one retracted, corrected, or questionable paper.

Of the 375 retracted papers surveyed, 76% continued to be cited after retraction, accumulating 9,662 citations—a long-standing disconnect between retraction notices and subsequent citations.

The citation illusion rate of 13 mainstream LLMs ranges from 14.23% to 94.93%; after analyzing 22 million citations from 2020 to 2025, invalid citations surged by 80.9% year-on-year in 2025.



One hundred verified forged citations passed the review of 3–5 experts at NeurIPS 2025; all were “composite deceptions” (a combination of real author, fake title, and real DOI).

A study in the field of mental health tested GPT-4o and found that 19.9% of the citations were completely fabricated; of the seemingly genuine citations, 45.4% contained bibliographical errors (most commonly invalid DOIs)—nearly two-thirds of the citations were either fabricated or contained errors.

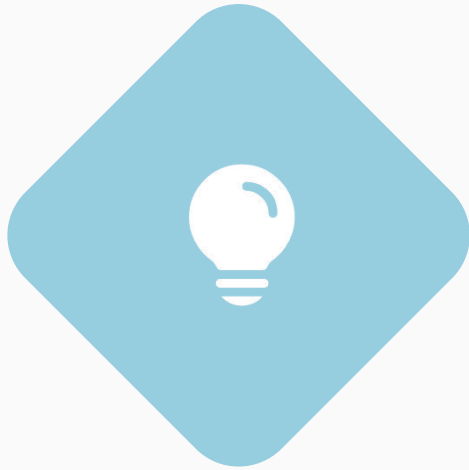
A GhostCite survey revealed that 76.7% of reviewers do not carefully check references, and 74.5% believe that peer review is ineffective in detecting citation errors.



# What exists today, and the gap

|                           | Core function                                                                                | Claim↔source alignment                                                              | In-writing integration               | Hallucination risk                   | Price / open-source                                                 | Key limitation                                                        |
|---------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------|--------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>scite.ai</b>           | Citation-context classification (support / contrast / mention) over 1.4B citation statements | ✗ Classifies the <i>context</i> around a citation, not <i>your</i> claim's accuracy | ✗ Browser extension, not in Overleaf | ✓ Grounded in real corpus            | \$20/mo, closed-source (acquired by Research Solutions for \$20.9M) | Macro sentiment, not sentence-level verification                      |
| <b>Semantic Scholar</b>   | Citation-graph search (280M nodes), TLDR summaries                                           | ✗ Distinguishes "influential vs. incidental" citations                              | ✗ Standalone search site             | ✓ Graph-based                        | Free                                                                | Doesn't enter the writing flow; weak on complex queries               |
| <b>Elicit</b>             | Semantic literature review, tabular extraction                                               | ✗ Discovery & synthesis, not in-writing verification                                | ✗ Separate web tool                  | ⚠ Inherits S2 corpus coverage gaps   | Freemium                                                            | Papers missing from S2 are invisible; biased toward empirical reviews |
| <b>Zotero</b>             | Open-source reference management + formatting                                                | ✗ Metadata & formatting only                                                        | ⚠ LaTeX via Better BibTeX            | ✓ Local data                         | Free, open-source                                                   | Verifies <i>format</i> , never <i>content accuracy</i>                |
| <b>Mendeley / EndNote</b> | Reference library + Word citation formatting                                                 | ✗ Same as above                                                                     | ✗ Word plugin, not Overleaf          | ✓ Local data                         | Mendeley free / EndNote \$275                                       | Mendeley privacy concerns; neither checks content                     |
| <b>ChatGPT / Claude</b>   | General chat + writing assistant                                                             | ✗ Generates "plausible-looking" references                                          | ✗ Generic chat UI                    | ✗ <b>High hallucination (14–95%)</b> | Subscription                                                        | Fabricates citations, no grounding, unauditable                       |
| ✦<br><b>ClaimTrace</b>    | Semantic lineage + sentence-level entailment verification                                    | ✓ <b>Core capability: claim ↔ source-passage alignment</b>                          | ✓ <b>Overleaf hover popup</b>        | ✓ Grounded in real PDFs              | Open-source, self-hostable                                          | New project — accuracy still being validated (our spike)              |

# The Market



## Target users at scale

- Overleaf: 25M+ users, 200M+ documents created, across 189 countries
- Target segment: researchers, PhD students, and co-author teams in CS / engineering / medicine



## Market size & growth

- Global reference-management software market: USD 371M in 2024 → USD 617M by 2031, a 7.6% CAGR



## Demand is already validated

- scite.ai was acquired by Research Solutions for \$20.9M, with 2M+ active users and 30+ publisher partnerships — proof that people pay for citation verification



## Our Wedge

- We don't compete on management (Zotero), citation counts (scite), or generation (ChatGPT).
- We own the "hover-to-verify" moment inside the writing workflow



# Solution Architecture

## 1. Research Workspace

Users access ClaimTrace through the Overleaf extension or the web dashboard. Source Processing

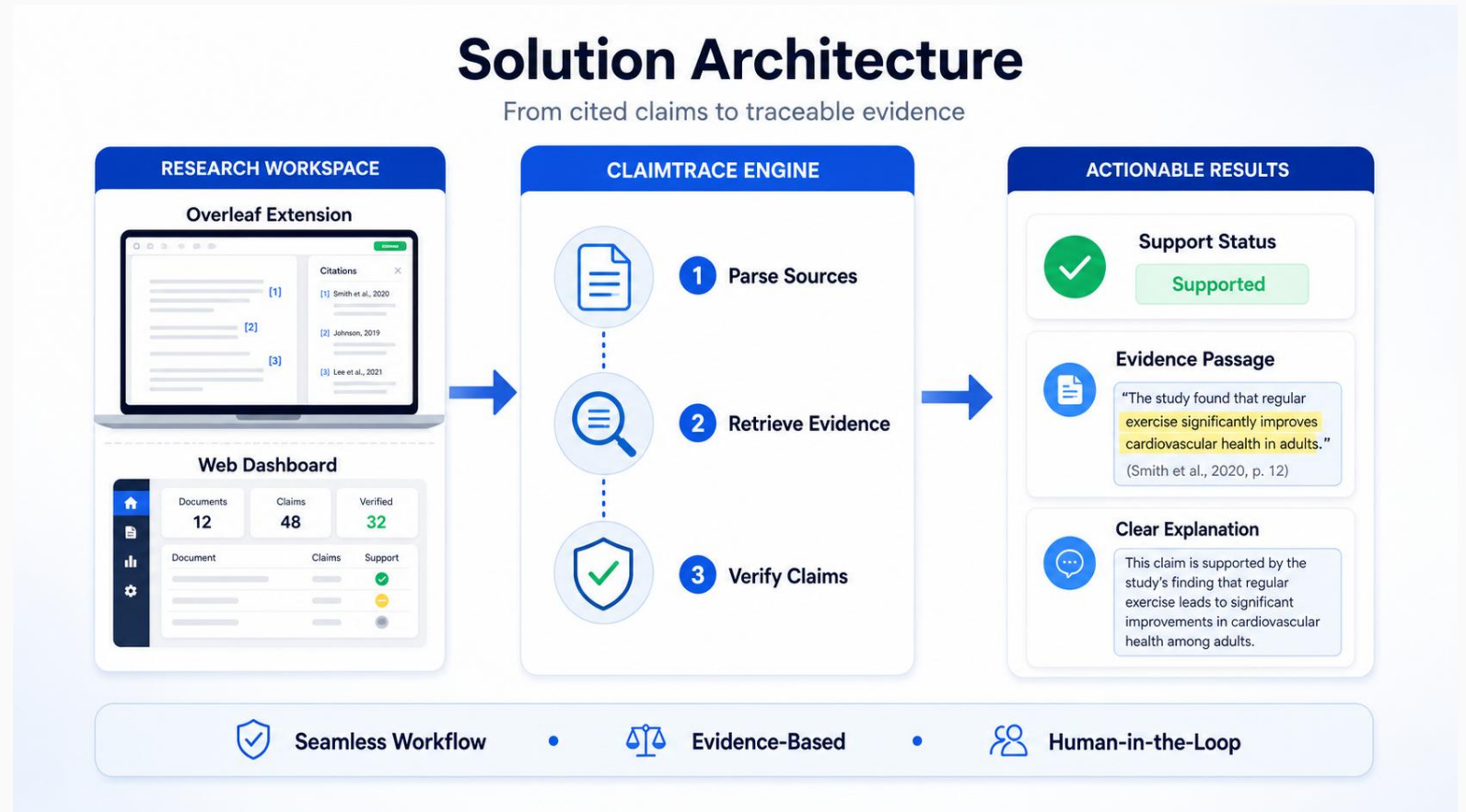
2. The system parses cited papers and extracts the information needed for verification.

## 3. Evidence & Verification

ClaimTrace retrieves relevant evidence passages and checks whether the claim is supported by the cited source.

## 4. Actionable Results

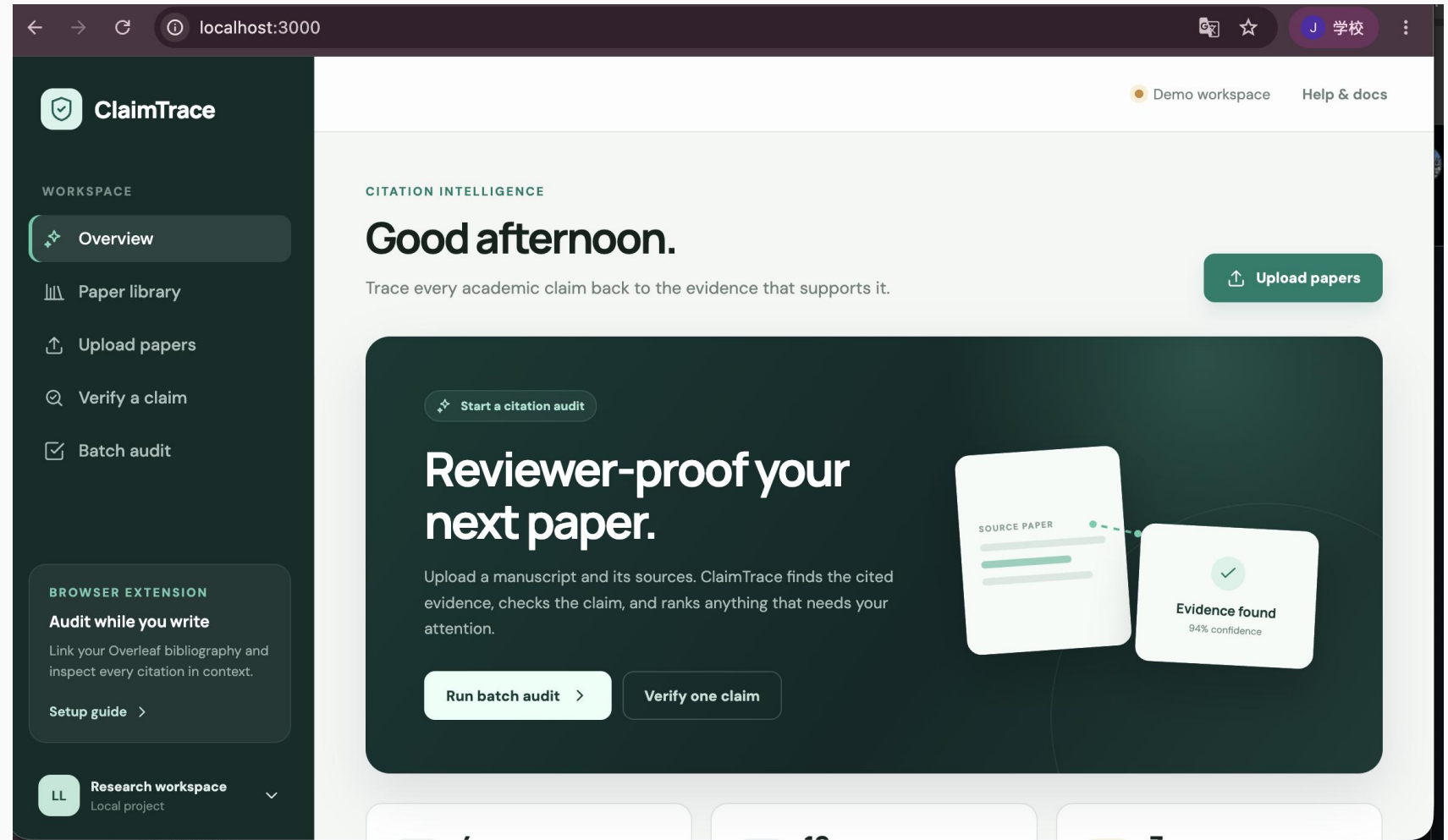
Users receive a support status, the matching evidence passage, and a clear explanation of the verification result.



# Web Dashboard

Users can upload papers, manage their paper library, verify a single claim, or run a batch audit.

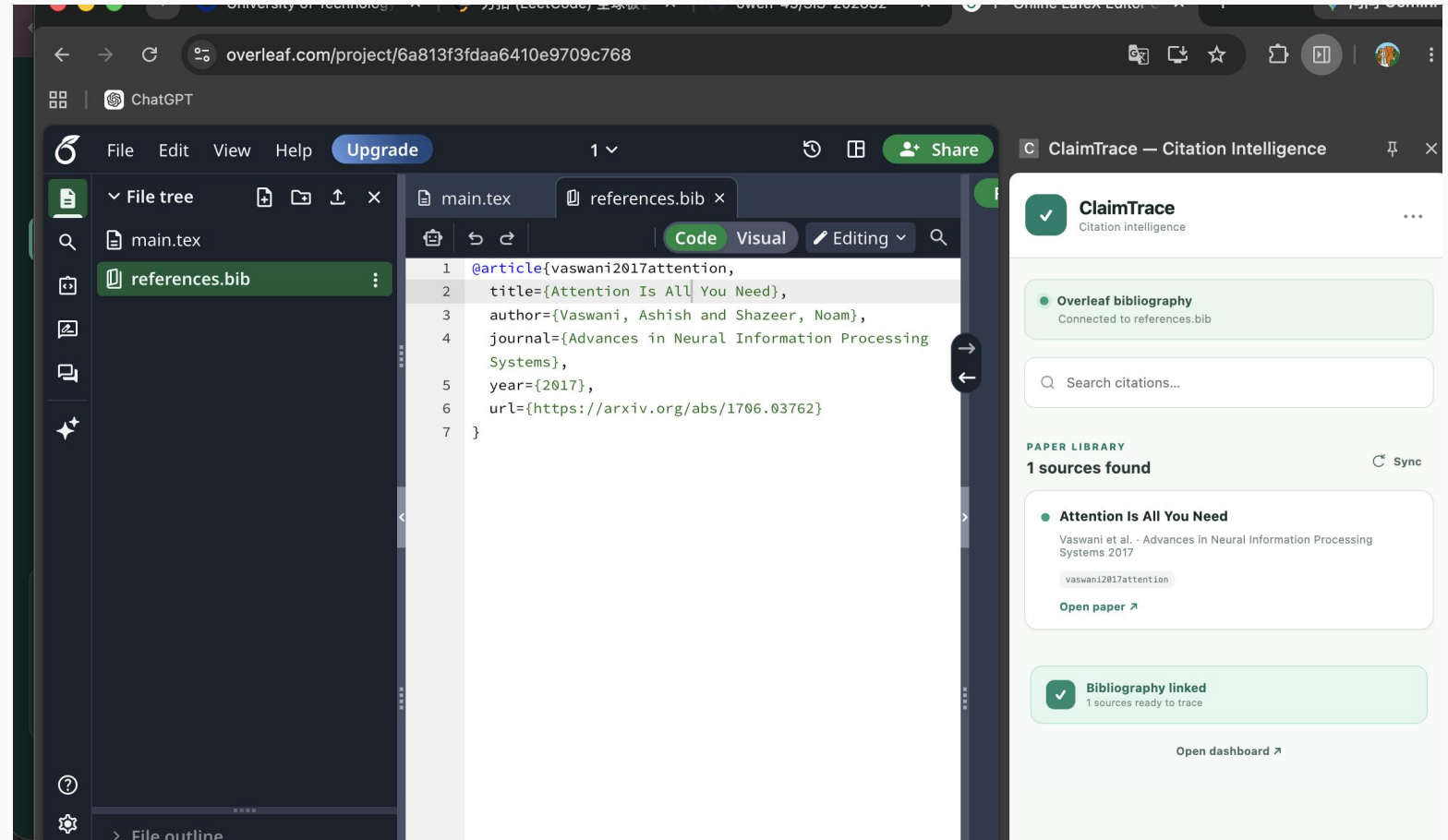
It provides one workspace for managing the citation verification process.



# Overleaf extension

It connects to the bibliography file and identifies the cited sources directly inside Overleaf.

This allows researchers to check citation evidence without leaving their writing environment.



## Check every citation before submission.

Upload your paper to find citation errors, duplicates, and missing sources.



Drag & drop your research paper here

PDF, DOCX, LaTeX up to 50 MB

Browse files

Private by design · Files encrypted · Nothing shared





overleaf.com/project/research-paper

Overleaf Research Paper

Recompile

PROJECT FILES

Research Paper

main.tex

references.bib

figures

references.bib

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1 @article{vaswani2017attention,
2   title={Attention Is All You Need},
3   author={Vaswani, Ashish and Shazeer, Noam and ...},
4   journal={Advances in Neural Information Processing Systems},
5   year={2017}, url={https://arxiv.org/abs/1706.03762}
6 }
7
8 @inproceedings{devlin2019bert,
9   title={BERT: Pre-training of Deep Bidirectional Transformers},
10  author={Devlin, Jacob and Chang, Ming-Wei and ...},
11  booktitle={NAACL}, year={2019}
12 }
13
14 @article{brown2020language,
15   title={Language Models are Few-Shot Learners},
16   author={Brown, Tom B. and Mann, Benjamin and ...},
17   year={2020}
18 }
```

Turn your .bib file into a paper library



The Team

# The Team

## Sichen Liu

- The team leader and the leader of the parser function
- Skills: Full-stack, RAG information retrieval

## HongYang Chen

- The leader of backend team
- Skills: Python, FastAPI, REST APIs, backend testing

## JunLi Li

- The leader of frontend team
- Skills: React, TypeScript, Vite, React Router, HTML/CSS, Responsive Web Design

## SiYuan Sun

- The member of backend team
- Skills: Python, REST APIs

## Yi Jiang

- The member of parser function team
- Skills: Multimodal analysis, Text recognition

## YiYang Yuan

- An engineer of frontend team, responsible for plugin design.
- skill: JavaScript, HTML/CSS

## Zheng Fu

The member of parser function team

Skill: Python, Web Design



T h a n k s   f o r  
w a t c h i n g



- <https://manusights.com/tools/reference-integrity/retracted-citation-study>
- [https://econpapers.repec.org/article/sprscient/v\\_3a130\\_3ay\\_3a2025\\_3ai\\_3a7\\_3ad\\_3a10.1007\\_5fs11192-025-05343-w.htm](https://econpapers.repec.org/article/sprscient/v_3a130_3ay_3a2025_3ai_3a7_3ad_3a10.1007_5fs11192-025-05343-w.htm)
- <https://libcat.scu.edu/EdsRecord/edsarx,edsarx.2602.06718>
- <https://arxiv-org.ezproxy.obspm.fr/abs/2602.05930>
- <https://libcat.scu.edu/EdsRecord/edsarx,edsarx.2602.06718>
- <https://www.digital-science.com/products/overleaf/>
- <https://www.marketresearch.com/GlobalInfoResearch-v4117/Global-Reference-Management-Software-Company-41646031/>